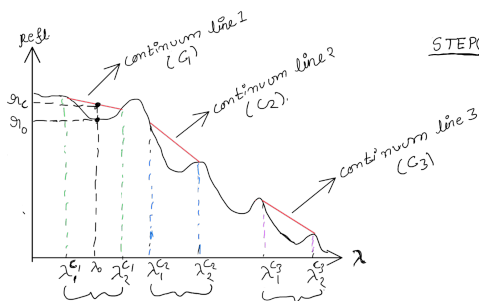




STEP 1

continuum removal formula is:

$$CR = \frac{R_0}{R_C} = \frac{\text{Original reflectance}}{\text{Continuum line value}} \text{ at certain wavelength.}$$

So, to find continuum removed reflectance at each wavelength within 3 regions first we need to find the continuum line & use it to find the denominator in above equation. (details in ②)

STEP 2

For this Region (say R1)
(540nm - 750nm)

We can define continuum line using simple slope & intercept:

$$\text{i.e. slope } (m^{C1}) = \frac{y_2^{C1} - y_1^{C1}}{\lambda_2^{C1} - \lambda_1^{C1}} = \frac{R_2 - R_1}{750 - 540}$$

& Intercept (b^{C1}) = $y_1^{C1} - (\text{slope}) \times \lambda_1^{C1}$ { this is from $y = mx + b$, we know $y_1^{C1}, \lambda_1^{C1} \neq m^{C1}$, so we use these to determine b }

$$\text{or, } b^{C1} = y_1^{C1} - m^{C1} \times \lambda_1^{C1}$$

So, using these slope & Intercept value we can find any value in continuum line corresponding to a wavelength.

For example: Let's say we want to determine continuum removed value at " λ_0 " indicated in figure. For that:

i) first we determine the value in continuum line corresponding to λ_0

using:

$$C_{\lambda_0} = \text{slope} \times \lambda_0 + \text{Intercept}$$

or, using our values:

$$C_{\lambda_0} = m^{C1} \times \lambda_0 + b^{C1} = R_C \text{ (labelled in figure)}$$

ii) Take the ratio with original reflectance as indicated in ①.

$$\text{i.e. } CR_{\lambda_0} = \frac{R_0}{C_{\lambda_0}} = \frac{\text{original reflectance at } \lambda_0}{\text{continuum removed reflectance at } \lambda_0}$$

If we repeat this step for each wavelength ($\lambda_1, \dots, \lambda_n$) we get continuum removed spectrum in Region 1 (R1). We have to do same for other 2 wavelength regions to get full continuum removed spectrum.